Exponent-Lattice Moment Masks for the Alaoglu–Erdős Conjecture

Toric Collapse, Stable Structural Support, Shared-Pole Newton Faces, and the Reduced-Denominator Endgame

A nonduplicative progress report continuing the supplied unified manuscript

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Bottom line. No unconditional proof is claimed. The new progress is a self-contained arithmetic framework for a class of auxiliary constructions not developed in the supplied report: integer "moment masks" in exponent-lattice near-unit coordinates. The framework produces five exact conclusions.

First, a collection of arbitrarily many continued-fraction near-relations has only two formal degrees of freedom: after quotienting by its toric lattice ideal, every mask is a Laurent polynomial in two adjacent near-units. Continued-fraction recurrences themselves therefore produce identically zero masks at both the input and output, not small nonzero integers.

Second, every one-direction Prouhet or Thue–Morse cancellation of order r clears to an integer divisible by the rth power of the corresponding output gap. High contact on one ray is thus not merely expensive; it algebraically factorizes into a large integer gap.

Third, a bivariate mask of contact order r clears into the ideal generated by the two output gaps to the rth power. An exact primewise Newton functional gives the guaranteed integer divisor and a product-formula contradiction criterion.

Fourth, the private-prime leading-face law is exact but not asymptotically decisive. For adjacent continued-fraction near-units, every structural prime has an eventually fixed sign: the reduced numerator and denominator supports stabilize, and both denominators are supported on the same nonempty set of primes. Their pole depths grow linearly and satisfy exact determinant-one transport identities.

Fifth, at each shared denominator prime the cleared mask is governed by an explicit two-dimensional Newton polygon. A unique minimizing monomial survives exactly; cancellation requires a tied initial form to vanish, and deep compression requires this to repeat through successive p-adic layers at every stable denominator prime. Together with effective two-logarithm separation, this proves that visible denominator clearing cannot become subunit and isolates the only surviving mechanism: near-total *reduced-denominator cancellation* through simultaneous, prime-specific shared Newton faces. The report ends with an exact sufficient criterion, a finite shared-face capture target, and a Lean-oriented dependency graph.

Prepared by OpenAI GPT-5.6 Pro.

Contents

1 Scope, nonduplication boundary, and evidence convention

The supplied unified report is exceptionally broad. It already develops the classical Four Exponentials reduction; the exact conditional solution monoid; valuation lattices, Kummer theory and local common exponents; generalized Vandermonde and Loewner determinants; finite differences, Newton interpolation and Smith profiles; Sturmian and carry models; Mahler and special-function transfers; canonical products and Stieltjes moments; rational point counting; support theorems; and multiple exact failure analyses. Repeating any of those surveys would add volume rather than information.

This supplement therefore imposes the following boundary.

- The exact conditional classification, finite exclusion ranges, Four Exponentials barrier, and all named results of the unified report are treated as inherited context and are not reproved.
- The new variables are *near-units attached to exponent-lattice relations*, not the original smooth nodes. The auxiliary objects are integer masks in those variables.
- Determinants appear only as the 2×2 lattice determinant controlling two independent near-relations. No new Vandermonde or interpolation determinant is proposed.
- The main arithmetic object is a single cleared rational value and its reduced denominator. This is intentionally scalar.
- Every assertion is marked as proved here, an external input, an exact computation, or an open target. The conjecture remains open at the end.

The public-claims audit in section 14 confirms that several recent machine-assisted mathematical results mentioned in the prompt have public records, but it found no public technical statement of the alleged new Alaoglu–Erdős step. Nothing from that unavailable claim is used below.

Status: **[proved here]** for the mathematical statements proved in this report; **[external theorem]** for explicitly cited inputs; **[exact computation]** for the small exact symbolic and continued-fraction computations; **[open target]** for the proposed endgame.

2 Inherited setup and the relation lattice

Write

$$\vartheta = \frac{\log 3}{\log 2}.$$

Suppose, for contradiction, that a nonintegral real number x satisfies

$$M := 2^x \in \mathbb{N}, \qquad N := 3^x \in \mathbb{N}. \tag{1}$$

The elementary interval argument gives $x > 1$: for $x < 0$ one has $0 < 2^x < 1$, and for $0 < x < 1$ one has $1 < 2^x < 2$. The supplied report proves much more, including transcendence of x and an exact primitive-generator theorem, but only the following elementary fact is needed here.

Lemma 2.1 (Output multiplicative independence)**.** *The integers M and N are multiplicatively independent.*

Proof. If $M^aN^b = 1$ for $a, b \in \mathbb{Z}$, then taking real logarithms and using (1) gives

$$x(a \log 2 + b \log 3) = 0.$$

Since $x \neq 0$, unique factorization applied to $2^a3^b = 1$ gives $a = b = 0$. □

For an exponent vector $\nu = (a, b) \in \mathbb{Z}^2$, define its input and output characters

$$\rho_\nu = 2^a3^b, \qquad \sigma_\nu = M^aN^b = \rho_\nu^x, \qquad \delta_\nu = a \log 2 + b \log 3. \tag{2}$$

Thus $\rho_\nu = e^{\delta_\nu}$ and $\sigma_\nu = e^{x\delta_\nu}$. A good rational approximation to ϑ supplies a nonzero ν for which δ_ν is small; then both ρ_ν and the rational number σ_ν are close to 1.

If p/q is a convergent to ϑ, the natural vector is

$$\nu = (-p, q), \qquad \delta_\nu = q \log 3 - p \log 2. \tag{3}$$

The central question of this report is whether several such rational output near-units can be combined by an integer polynomial mask to produce a nonzero rational number that is too small for its reduced denominator.

2.1 Exact reduced-height growth

The denominator cost of a rational output near-unit is controlled by the valuation geometry of (M, N), independently of continued fractions.

Let $\mathcal{P}$ be the finite set of primes dividing MN, and put

$$\mathbf{m} = (v_p(M))_{p\in\mathcal{P}}, \qquad \mathbf{n} = (v_p(N))_{p\in\mathcal{P}}.$$

For $\nu = (a, b)$ define the weighted valuation norm

$$L_{M,N}(\nu) = \sum_{p\in\mathcal{P}} |av_p(M) + bv_p(N)| \log p. \tag{4}$$

Lemma 2.2 (Valuation norm)**.** *The function $L_{M,N}$ extends to a norm on $\mathbb{R}^2$. Consequently there is a constant $c_{M,N} > 0$ such that*

$$L_{M,N}(a, b) \geq c_{M,N}(|a| + |b|) \qquad (a, b \in \mathbb{R}). \tag{5}$$

Proof. It is a seminorm by construction. If it vanishes at (a, b), then $a\mathbf{m} + b\mathbf{n} = 0$. For rational (a, b) this would give a multiplicative relation between M and N; hence the two valuation vectors are linearly independent over $\mathbb{Q}$. Because they have integral coordinates, they are also linearly independent over $\mathbb{R}$. Thus the seminorm has trivial kernel. Norm equivalence in the finite-dimensional space $\mathbb{R}^2$ gives (5). □

Write the rational number σ_ν in lowest terms as

$$\sigma_\nu = \frac{R_\nu}{S_\nu}, \qquad R_\nu, S_\nu \in \mathbb{N}, \qquad \gcd(R_\nu, S_\nu) = 1.$$

Proposition 2.3 (Exact numerator–denominator law)**.** *For every $\nu \in \mathbb{Z}^2$,*

$$\log R_\nu + \log S_\nu = L_{M,N}(\nu), \tag{6}$$
$$\log R_\nu - \log S_\nu = x\delta_\nu. \tag{7}$$

In particular,

$$h(\sigma_\nu) := \log \max(R_\nu, S_\nu) = \frac{L_{M,N}(\nu) + |x\delta_\nu|}{2}, \tag{8}$$

and if $|x\delta_\nu| \le 1$, then

$$\min(\log R_\nu, \log S_\nu) \ge \frac{c_{M,N}(|a| + |b|) - 1}{2}. \tag{9}$$

Proof. The exponent of a prime p in σ_ν is $e_p = av_p(M) + bv_p(N)$. Reduction separates the positive and negative parts:

$$\log R_\nu = \sum_p \max(e_p, 0) \log p, \qquad \log S_\nu = \sum_p \max(-e_p, 0) \log p.$$

Their sum is (6); their difference is $\log \sigma_\nu = x\delta_\nu$, giving (7). The final two formulas follow by adding and subtracting. □

The significance is quantitative. A nontrivial rational output near-unit can be archimedeanly close to 1, but its reduced numerator and denominator are both exponentially large in the exponent-vector length. Any auxiliary mask must defeat that height, not merely make a Taylor series start late.

2.2 A two-direction uncertainty identity

Let

$$\nu = (a, b), \qquad \omega = (c, d), \qquad D = \det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc.$$

Proposition 2.4 (Exact relation-lattice uncertainty)**.** *One has*

$$d\delta_\nu - b\delta_\omega = D \log 2, \qquad a\delta_\omega - c\delta_\nu = D \log 3. \tag{10}$$

Consequently, if $D \neq 0$ and $H = \|\nu\|_1 + \|\omega\|_1$, then

$$\max(|\delta_\nu|, |\delta_\omega|) \ge \frac{|D| \max(\log 2, \log 3)}{H}. \tag{11}$$

Proof. Substitute $\delta_\nu = a \log 2 + b \log 3$ and $\delta_\omega = c \log 2 + d \log 3$. The cross terms cancel and give (10). Each right-hand side is bounded above in absolute value by $H \max(|\delta_\nu|, |\delta_\omega|)$. □

For adjacent convergents $p_n/q_n, p_{n+1}/q_{n+1}$ to ϑ, the vectors $\nu_n = (-p_n, q_n)$ and $\nu_{n+1} = (-p_{n+1}, q_{n+1})$ have determinant ± 1. They are therefore the optimally conditioned integral basis for two-direction near-unit constructions. Formula (11) says that even this basis cannot make both logarithmic errors smaller than the reciprocal exponent scale.

Status: [proved here] for theorems 2.1 to 2.4. These are elementary and independent of the deep transcendence inputs in the unified report.

3 Toric reduction: many near-relations have only two formal degrees of freedom

A natural reaction to the two-direction uncertainty bound is to use many convergents and Prouhet-type cancellations among them. The first new structural result shows exactly what additional variables do and do not buy.

Fix exponent vectors $\nu_i = (a_i, b_i) \in \mathbb{Z}^2$ for $1 \le i \le k$. Define the formal monomial map

$$\Psi_\nu : \mathbb{Z}[T_1^{\pm 1}, \dots, T_k^{\pm 1}] \longrightarrow \mathbb{Z}[U^{\pm 1}, V^{\pm 1}], \qquad T_i \longmapsto U^{a_i} V^{b_i}. \tag{12}$$

For $\alpha = (\alpha_1, \dots, \alpha_k) \in \mathbb{Z}^k$ write $T^\alpha = \prod_i T_i^{\alpha_i}$ and

$$\pi_\nu(\alpha) = \sum_i \alpha_i \nu_i \in \mathbb{Z}^2.$$

Definition 3.1 (Toric lattice ideal)**.** The lattice ideal of the relation list ν is

$$I_\nu = \left\langle T^\alpha - T^\beta : \pi_\nu(\alpha) = \pi_\nu(\beta) \right\rangle \subset \mathbb{Z}[T_1^{\pm 1}, \dots, T_k^{\pm 1}]. \tag{13}$$

Theorem 3.2 (Universal toric kernel)**.** *The kernel of Ψ_ν is exactly I_ν.*

Proof. Every displayed binomial has the same aggregate exponent on its two terms, so $I_\nu \subseteq \ker \Psi_\nu$. Conversely, let $C = \sum_\alpha c_\alpha T^\alpha$ lie in the kernel. Group its terms by the fibers of π_ν. Laurent monomials $U^r V^s$ are a free $\mathbb{Z}$-basis of the target ring, so in each fiber the sum of the coefficients is zero. Choose one reference exponent α_0 in a fiber. Then

$$\sum_{\alpha \text{ in the fiber}} c_\alpha T^\alpha = \sum_{\alpha \neq \alpha_0} c_\alpha (T^\alpha - T^{\alpha_0}),$$

because the coefficient of T^{α_0} is the negative sum of the others. Every difference belongs to I_ν. $\square$

Corollary 3.3 (Universal identities transfer to the output)**.** *If $C \in I_\nu$, then*

$$C(\rho_{\nu_1}, \dots, \rho_{\nu_k}) = 0, \qquad C(\sigma_{\nu_1}, \dots, \sigma_{\nu_k}) = 0. \tag{14}$$

Proof. Compose (12) first with $(U, V) = (2, 3)$ and then with $(U, V) = (M, N)$. $\square$

Warning 3.4 (The theorem is formal, not a classification of numerical additive identities)**.** The numerical evaluation kernel at $(2, 3)$ is larger than I_ν, because $3 - 2 - 1 = 0$ and similar additive identities exist. The theorem classifies identities caused solely by integer dependencies of exponent vectors. Such identities are precisely the ones that transfer automatically to every pair of bases. Candidate-specific additive cancellation is not ruled out; it is the difficult part.

3.1 Unimodular two-variable normal form

Theorem 3.5 (Two-variable normal form)**.** *Suppose two vectors ν_1, ν_2 form a unimodular basis of $\mathbb{Z}^2$. For every $i \ge 3$ there are unique integers r_i, s_i such that $\nu_i = r_i \nu_1 + s_i \nu_2$. Modulo I_ν, every mask has a unique Laurent normal form*

$$F(U, V) \in \mathbb{Z}[U^{\pm 1}, V^{\pm 1}], \qquad T_1 \mapsto U, \quad T_2 \mapsto V, \quad T_i \mapsto U^{r_i} V^{s_i}. \tag{15}$$

In particular, adding arbitrarily many convergent variables does not increase the formal dimension beyond two.

Proof. Unimodularity makes (ν_1, ν_2) a $\mathbb{Z}$-basis, giving the unique coordinates. The monomial map is then surjective onto a Laurent polynomial ring in the two basis monomials, and theorem 3.2 identifies its kernel. □

Example 3.6 (Continued-fraction recurrence masks are identically zero)**.** Continued-fraction vectors satisfy

$$\nu_{n+1} = a_{n+1}\nu_n + \nu_{n-1}.$$

Therefore

$$T_{n+1} - T_n^{a_{n+1}} T_{n-1} \in I_\nu. \tag{16}$$

The corresponding input and output identities are exact. Iterating the recurrence can create impressive formal cancellation, but every such recurrence-only mask evaluates to zero rather than to a useful nonzero small integer.

This is the first principal no-go result of the report. A many-convergent construction must be reduced modulo I_ν before its contact order is credited. Otherwise torically trivial zeros can be mistaken for high-order arithmetic cancellation.

Status: [proved here] for the universal toric kernel, its transfer to both base pairs, and the unimodular normal form. The result is an elementary special case of the theory of lattice and binomial ideals, included with a complete proof so that no black box is needed.

4 Moment masks and contact order at the identity

After toric reduction, the right object is a Laurent polynomial in two near-unit variables. Multiplying by a monomial changes neither vanishing order at $(1, 1)$ nor the essential arithmetic problem, so we may work with an ordinary polynomial

$$F(U, V) = \sum_{(i,j)\in\Lambda} c_{ij} U^i V^j \in \mathbb{Z}[U, V], \qquad c_{ij} \neq 0.$$

Definition 4.1 (Contact order)**.** The contact order of a nonzero mask F at the identity is

$$\operatorname{ord}_{(1,1)} F = \min\{r + s : [X^r Y^s]F(1 + X, 1 + Y) \neq 0\}. \tag{17}$$

Equivalently, it is the largest integer m such that $F \in (U - 1, V - 1)^m$ in $\mathbb{Z}[U, V]$.

If the contact order is r, evaluation at $U = e^{\varepsilon_1}$, $V = e^{\varepsilon_2}$ begins at total degree r in the two small logarithmic errors. The terminology "moment mask" comes from the following exact characterization.

Proposition 4.2 (Two-dimensional Prouhet moment equations)**.** *Let* $F(U, V) = \sum_{\ell=1}^{s} c_\ell U^{a_\ell} V^{b_\ell}$ *with distinct exponent points* (a_ℓ, b_ℓ)*. The following are equivalent for an integer* $r \geq 1$*.*

1. $\operatorname{ord}_{(1,1)} F \geq r$.
2. *For all* $i, j \geq 0$ *with* $i + j < r$,

$$\sum_{\ell=1}^{s} c_\ell \, (a_\ell)_{\underline{i}} \, (b_\ell)_{\underline{j}} = 0. \tag{18}$$

3. For all $i, j \geq 0$ with $i + j < r$,

$$\sum_{\ell=1}^{s} c_\ell a_\ell^i b_\ell^j = 0. \tag{19}$$

Proof. The mixed derivative $\partial_U^i \partial_V^j F(1,1)$ is the left side of (18). Vanishing of all derivatives of total order below r is precisely membership in $(U-1, V-1)^r$. Falling factorials and ordinary powers are related by unitriangular Stirling transforms in each variable, so the two families of moment equations are equivalent. □

Thus a contact-r mask is a signed, weighted two-dimensional Prouhet–Tarry–Escott configuration on the exponent lattice. The next bounds quantify the unavoidable size of such a configuration.

Theorem 4.3 (Degree and support ceilings)**.** *Let $F \neq 0$ have bidegree (d, e) and s nonzero monomials. Then*

$$\operatorname{ord}_{(1,1)} F \leq d + e, \qquad \operatorname{ord}_{(1,1)} F \leq s - 1. \tag{20}$$

Proof. The polynomial $F(1 + X, 1 + Y)$ has total degree at most $d + e$, giving the first bound. For the second, write the exponent points as (a_ℓ, b_ℓ). Choose real numbers α, β so that the frequencies $\lambda_\ell = \alpha a_\ell + \beta b_\ell$ are distinct and so that the first nonzero homogeneous part of $F(1 + X, 1 + Y)$ does not vanish on the tangent direction (α, β). Then

$$g(t) = F(e^{\alpha t}, e^{\beta t}) = \sum_{\ell=1}^{s} c_\ell e^{\lambda_\ell t}$$

has a zero at 0 of exactly the same order as F at $(1,1)$. If that order were at least s, then $g^{(m)}(0) = \sum_\ell c_\ell \lambda_\ell^m = 0$ for $0 \leq m < s$. The Vandermonde matrix on the distinct λ_ℓ is invertible, forcing every $c_\ell = 0$, a contradiction. □

The support ceiling is a useful audit rule. A claimed order-r mask with at most r monomials is either wrong or torically zero after normalization.

4.1 A uniform local estimate

Write

$$\widetilde{F}(X, Y) = F(1 + X, 1 + Y) = \sum_{i,j} \gamma_{ij} X^i Y^j, \qquad \|\widetilde{F}\|_1 = \sum_{i,j} |\gamma_{ij}|. \tag{21}$$

Lemma 4.4 (Contact estimate)**.** *Suppose* $\operatorname{ord}_{(1,1)} F \geq r$ *and* $|\varepsilon_1|, |\varepsilon_2| \leq \frac{1}{4}$*. Put* $\Delta = \max(|\varepsilon_1|, |\varepsilon_2|)$*. Then*

$$|F(e^{\varepsilon_1}, e^{\varepsilon_2})| \leq \|\widetilde{F}\|_1 (2\Delta)^r. \tag{22}$$

Proof. For $|t| \leq 1/4$, one has $|e^t - 1| \leq 2|t|$. Every nonzero term in (21) has $i + j \geq r$, and $2\Delta \leq 1$, so $|e^{\varepsilon_1} - 1|^i |e^{\varepsilon_2} - 1|^j \leq (2\Delta)^r$. Sum the absolute values. □

The estimate is deliberately crude but universal. Later we prove that this entire style of estimate, when followed by full denominator clearing, cannot close the conjecture on adjacent convergents. Its failure identifies reduced-denominator cancellation as the only remaining scalar-mask mechanism.

5 One-direction Prouhet cancellation is an exact gap power

Before using two variables, it is important to close the seemingly attractive one-variable route completely.

Theorem 5.1 (One-direction gap-power theorem). *Let $P(T) \in \mathbb{Z}[T]$ have degree d and a zero of multiplicity at least r at $T = 1$. For integers R, S with $S \neq 0$, define*

$$I_P(R, S) = S^d P(R/S) \in \mathbb{Z}. \tag{23}$$

Then

$$(R - S)^r \mid I_P(R, S). \tag{24}$$

If $P(R/S) \neq 0$, then

$$|I_P(R, S)| \geq |R - S|^r. \tag{25}$$

Proof. Because $T - 1$ is monic, repeated integral polynomial division gives $P(T) = (T - 1)^r Q(T)$ with $Q \in \mathbb{Z}[T]$ and $\deg Q = d - r$. Hence

$$S^d P(R/S) = (R - S)^r S^{d-r} Q(R/S),$$

and the final factor is an integer. The lower bound follows if the product is nonzero. □

This theorem covers every one-dimensional Prouhet partition, finite-difference mask, and Thue–Morse sign pattern at once. Coefficient cancellation cannot make the cleared integer smaller than the corresponding output gap power.

5.1 The Thue–Morse calibration

For $r \geq 1$, define

$$P_r(T) = \prod_{j=0}^{r-1} (1 - T^{2^j}) = \sum_{n=0}^{2^r - 1} (-1)^{s_2(n)} T^n, \tag{26}$$

where $s_2(n)$ is the sum of the binary digits of n. It has degree $2^r - 1$, exactly 2^r coefficients of absolute value 1, and a zero of exact multiplicity r at 1. The corresponding Prouhet identity equalizes power sums of degrees $0, \ldots, r - 1$ between the even and odd Thue–Morse classes.

Corollary 5.2 (Exact Thue–Morse factorization). *For $d = 2^r - 1$,*

$$S^d P_r(R/S) = \prod_{j=0}^{r-1} (S^{2^j} - R^{2^j}). \tag{27}$$

Every factor on the right is divisible by $S - R$, so the cleared value is divisible by $(S - R)^r$.

Proof. The exponents 2^j sum to $2^r - 1$, so denominator clearing distributes exactly across the product in (26). □

The high contact is real, but the arithmetic output becomes a product of increasingly large integer differences. This is the opposite of a subunit certificate.

Table 1: Exact size data for the first Thue–Morse masks.

contact r	degree 2^r-1	coefficient ℓ^1 norm	multiplicity at 1
1	1	2	1
2	3	4	2
3	7	8	3
4	15	16	4
5	31	32	5
6	63	64	6
7	127	128	7

Closed route. No mask supported on integer powers of a single rational output near-unit can prove the conjecture by "high Prouhet contact plus denominator clearing." If the cleared value is nonzero, its exact gap-power divisor grows with the contact order.

6 Bivariate clearing: the gap ideal and its primewise Newton functional

Two independent near-unit directions can distribute contact among two gaps and thereby evade the individual gap power in theorem 5.1. The replacement is an exact ideal-membership statement.

Let

$$z_1 = \frac{R_1}{S_1}, \qquad z_2 = \frac{R_2}{S_2}, \qquad G_1 = R_1 - S_1, \qquad G_2 = R_2 - S_2,$$

where at first no coprimality assumption is needed. Let $F \in \mathbb{Z}[U, V]$ have bidegree at most (d, e) and shifted expansion

$$F(1 + X, 1 + Y) = \sum_{\substack{0 \le i \le d \\ 0 \le j \le e}} \gamma_{ij} X^i Y^j. \tag{28}$$

Define the fully cleared integer

$$J_F(z_1, z_2) = S_1^d S_2^e F(z_1, z_2) \in \mathbb{Z}. \tag{29}$$

Theorem 6.1 (Gap-ideal theorem)**.** *If* $\operatorname{ord}_{(1,1)} F \ge r$*, then*

$$J_F(z_1, z_2) = \sum_{i+j \ge r} \gamma_{ij} G_1^i G_2^j S_1^{d-i} S_2^{e-j} \in (G_1, G_2)^r \subset \mathbb{Z}. \tag{30}$$

Consequently,

$$\gcd(G_1, G_2)^r \mid J_F(z_1, z_2). \tag{31}$$

Proof. Substitute $X = G_1/S_1$ and $Y = G_2/S_2$ in (28) and multiply by $S_1^d S_2^e$. Contact order removes all terms with $i + j < r$. Every remaining monomial lies in $(G_1, G_2)^r$, and every one is divisible by the rth power of their integer gcd. □

The gcd lower bound is often far from the full ideal information. Prime by prime, the support of the shifted mask determines a sharper guaranteed divisor.

Definition 6.2 (Primewise Newton floor)**.** For a prime p, write

$$g_{k,p} = v_p(G_k), \qquad s_{k,p} = v_p(S_k) \quad (k = 1, 2).$$

Define

$$\tau_p(F; z_1, z_2) = \min_{\gamma_{ij} \neq 0} \big(v_p(\gamma_{ij}) + i g_{1,p} + j g_{2,p} + (d - i) s_{1,p} + (e - j) s_{2,p} \big). \tag{32}$$

Proposition 6.3 (Exact guaranteed divisor)**.** *For every prime p,*

$$v_p(J_F(z_1, z_2)) \geq \tau_p(F; z_1, z_2). \tag{33}$$

Thus the finite product

$$\mathcal{D}_F(z_1, z_2) = \prod_p p^{\tau_p(F; z_1, z_2)} \tag{34}$$

divides $J_F(z_1, z_2)$.

Proof. Each term of (30) has the valuation displayed inside the minimum. The valuation of a sum is at least the minimum valuation of its summands. ☐

The functional in (32) is the lower support function of the shifted Newton polygon evaluated against a prime-dependent weight vector. Unlike a determinant valuation, it is a scalar certificate and can be optimized directly over the mask support.

Criterion 6.4 (Product-formula mask criterion)**.** *Suppose $F(z_1, z_2) \neq 0$. Then*

$$\mathcal{D}_F(z_1, z_2) \leq |J_F(z_1, z_2)| \leq \sum_{i,j} |\gamma_{ij}| |G_1|^i |G_2|^j S_1^{d-i} S_2^{e-j}. \tag{35}$$

Therefore a contradiction follows if the explicit right-hand side is strictly smaller than $\mathcal{D}_F(z_1, z_2)$.

Proof. The first inequality follows because the nonzero integer J_F is divisible by $\mathcal{D}_F$; the second is the triangle inequality applied to (30). ☐

Example 6.5 (A positive cross-face mask)**.** For $r \geq 1$,

$$Q_r(U, V) = (V + 1)(U - 1)^{2r} + (U + 1)(V - 1)^{2r} \tag{36}$$

has contact order $2r$ and is strictly positive for positive $(U, V) \neq (1, 1)$. For $r = 2$, full bidegree-$(4, 4)$ clearing gives the exact identity

$$S_1^4 S_2^4 Q_2(z_1, z_2) = (R_2 + S_2) S_2^3 G_1^4 + (R_1 + S_1) S_1^3 G_2^4. \tag{37}$$

This example guarantees nonvanishing without a determinant. It will nevertheless fail at private denominator primes unless the cross-face values $z_2 + 1$ and $z_1 + 1$ provide the necessary divisibility; section 7 makes that statement exact.

Status: [proved here] for the gap-ideal theorem, the primewise Newton floor, and the scalar product-formula criterion. These statements are finite identities and valuation inequalities, with no conjectural input.

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7 Leading faces: the private-prime prototype

The full clearing factor $S_1^d S_2^e$ is usually much larger than the reduced denominator of $F(z_1, z_2)$, so a useful construction must force cancellation between that clearing factor and the cleared numerator. The next theorem identifies the first obstruction exactly.

Assume now that $z_1 = R_1/S_1$ and $z_2 = R_2/S_2$ are in lowest terms, and that $d \geq 1$. Write

$$F(U, V) = \sum_{i=0}^{d} f_i(V)U^i, \qquad f_i \in \mathbb{Z}[V], \qquad \deg f_i \leq e, \qquad f_d \neq 0. \tag{38}$$

The polynomial $f_d(V)$ is the top U-face of the Newton polygon. Put

$$A_i = S_2^e f_i(z_2) \in \mathbb{Z}. \tag{39}$$

Then

$$J_F = S_1^d S_2^e F(z_1, z_2) = \sum_{i=0}^{d} A_i R_1^i S_1^{d-i}. \tag{40}$$

Theorem 7.1 (Leading-face survival at a private prime)**.** *Let $p \mid S_1$ and $p \nmid S_2$, and put $s = v_p(S_1)$. If*

$$v_p(A_d) < s, \tag{41}$$

then

$$v_p(J_F) = v_p(A_d). \tag{42}$$

Consequently the exponent of p in the reduced denominator of $F(z_1, z_2)$ is exactly

$$ds - v_p(A_d). \tag{43}$$

In particular, if $p \nmid A_d$, the entire pole p^{ds} survives.

Proof. Since z_1 is reduced, $p \nmid R_1$. The $i = d$ term in (40) has valuation $v_p(A_d)$. Every term with $i < d$ has valuation at least $(d - i)s \geq s$, because A_i is an integer. Under (41), the top-face term is the unique term of minimal valuation, so the ultrametric inequality is an equality and gives (42). The full clearing denominator has p-valuation ds, and $p \nmid S_2$, proving (43). $\square$

Corollary 7.2 (Necessary first-layer capture)**.** *To reduce the p-part of the denominator below $p^{(d-1)s}$ at a prime satisfying $p \mid S_1$, $p \nmid S_2$, it is necessary that*

$$p^s \mid S_2^e f_d(z_2). \tag{44}$$

If the divisibility holds at every prime in the private part

$$P_1 = \prod_{p \mid S_1,\ p \nmid S_2} p^{v_p(S_1)}, \tag{45}$$

then either $f_d(z_2) = 0$ or

$$\log P_1 \leq eh(z_2) + \log \|f_d\|_1, \tag{46}$$

where $h(z_2) = \log \max(R_2, S_2)$ and $\|f_d\|_1$ is the coefficient ℓ^1 norm.

Proof. The first statement is the contrapositive of theorem 7.1. If it holds for all private primes, then $P_1 \mid A_d$. Unless $A_d = 0$,

$$P_1 \le |A_d| = |S_2^e f_d(R_2/S_2)| \le \|f_d\|_1 \max(R_2, S_2)^e,$$

which gives (46). $\square$

This is a genuinely asymmetric statement: cancellation of the top U-pole layer must be paid for by the value of the top U-face at the other near-unit. The symmetric theorem is obtained by exchanging U, V.

7.1 Monomial transport versus unit-root capture

Factor the top face as

$$f_d(V) = V^m \widehat{f}_d(V), \qquad \widehat{f}_d(0) \neq 0. \tag{47}$$

At a prime $p \nmid S_2$, the monomial V^m contributes $mv_p(R_2)$ to $v_p(A_d)$. This is deterministic transport of valuation from the numerator of z_2 to the denominator of z_1. It is exactly the min-plus or tropical mechanism already audited in the supplied report.

After that monomial contribution is removed, any additional valuation must come from the congruence

$$\widehat{f}_d(z_2) \equiv 0 \pmod{p^k} \tag{48}$$

when z_2 is a p-adic unit. We call this *unit-root capture*. It is the genuinely new denominator mechanism isolated by the mask framework.

Corollary 7.3 (Face-monic obstruction)**.** *Suppose the top U-face is the constant $f_d(V) = \pm 1$. Then at every prime $p \mid S_1$ with $p \nmid S_2$, the full $p^{dv_p(S_1)}$ denominator survives. More generally, if $f_d(V) = \pm V^m$ and $p \nmid R_2 S_2$, the full pole survives.*

Proof. In the first case $A_d = \pm S_2^e$; in the second it is a p-adic unit times a power of the p-adic unit z_2. Apply theorem 7.1. $\square$

Example 7.4 (Natural positive masks are face-monic)**.** The mask

$$F_r(U, V) = (U - 1)^{2r} + (V - 1)^{2r} \tag{49}$$

is positive for positive $(U, V) \neq (1, 1)$ and has contact order $2r$. Its top U-face and top V-face are both 1. Hence every denominator prime private to either near-unit survives with full multiplicity. Positivity solves nonvanishing but leaves the denominator tax untouched.

7.2 The pole-layer Newton polygon

The first-layer theorem is the first edge of a more complete local picture. At a prime $p \mid S_1$, $p \nmid S_2$, put $s = v_p(S_1)$ and consider the points

$$(i, v_p(A_i)), \qquad 0 \le i \le d. \tag{50}$$

The term with index i in (40) has valuation $v_p(A_i) + (d-i)s$. Thus $v_p(J_F)$ is at least the minimum of the affine functional

$$\ell_{p,s}(i,t) = t + (d-i)s \tag{51}$$

over the coefficient Newton polygon, with equality whenever the minimum is unique. Full cancellation of successive pole layers requires the right-hand vertices to rise by at least s per horizontal step, or requires exact cancellation among tied minimal terms. Either phenomenon translates to high p-adic divisibility of a short list of face evaluations.

Research target 7.5 (Iterated face-capture theorem)**.** Develop a global, coefficient-height-efficient bound for the total number of canceled pole layers in terms of the evaluated face chain

$$f_d(z_2), f_{d-1}(z_2), \ldots$$

and the resultants of successive faces. The desired form is a lower bound for the reduced denominator that remains valid even when several local minima tie and cancel.

The first-layer result is already enough to rule out all face-monic constructions and to turn cyclotomic faces into an explicit finite-order covering problem.

Status: [proved here] for leading-face survival, the private first-layer condition, its height budget, and the face-monic obstruction. Theorem 7.5 is open and is one of the two strongest concrete follow-up problems identified here.

8 Cyclotomic cross-faces: a finite private-prime prototype

A polynomial leading face can capture a unit-private prime only by having the other near-unit as a root modulo that prime. Cyclotomic polynomials give the most structured way to enforce such roots while retaining positivity on the positive real axis.

For a finite set $\mathcal{M} \subset \{2, 3, \ldots\}$ define

$$A_{\mathcal{M}}(T) = \prod_{m \in \mathcal{M}} \Phi_m(T), \qquad E(\mathcal{M}) = \deg A_{\mathcal{M}} = \sum_{m \in \mathcal{M}} \varphi(m). \tag{52}$$

Lemma 8.1 (Positivity and coefficient budget)**.** *For every $m \geq 2$, $\Phi_m(t) > 0$ for all $t > 0$. Moreover*

$$\|A_{\mathcal{M}}\|_1 \leq 2^{E(\mathcal{M})}. \tag{53}$$

Proof. The polynomial Φ_m has no positive real root for $m \geq 2$, has positive leading coefficient, and has value 1 at 0, so it is positive on $(0, \infty)$. Its roots all have modulus 1. Therefore the absolute value of its coefficient of degree j is at most the corresponding binomial coefficient, and $\|\Phi_m\|_1 \leq 2^{\varphi(m)}$. The ℓ^1 norm is submultiplicative under polynomial multiplication. □

For $m \geq 2$ define the homogenized cyclotomic polynomial by

$$\Phi_m^{\mathrm{hom}}(R, S) := S^{\varphi(m)} \Phi_m(R/S) \in \mathbb{Z}[R, S]. \tag{54}$$

Proposition 8.2 (Cyclotomic order detection). *Let $z = R/S$ be reduced and let $p \nmid RS \prod_{m\in\mathcal{M}} m$. Then*

$$p \mid S^{E(\mathcal{M})} A_{\mathcal{M}}(z) \quad \Longleftrightarrow \quad \mathrm{ord}_p(RS^{-1}) \in \mathcal{M}, \tag{55}$$

where the order is in $\mathbb{F}_p^\times$.

Proof. The denominator is a p-adic unit. For $p \nmid m$, an element of $\mathbb{F}_p^\times$ is a root of Φ_m if and only if it has exact multiplicative order m. The product vanishes precisely when one factor does. □

For finite order sets $\mathcal{M}, \mathcal{N}$ and $r \geq 1$, consider

$$\mathcal{F}_{r,\mathcal{M},\mathcal{N}}(U,V) = A_{\mathcal{M}}(V)(U-1)^{2r} + A_{\mathcal{N}}(U)(V-1)^{2r}. \tag{56}$$

Proposition 8.3 (Positive cyclotomic-face mask). *The mask in (56) is strictly positive for all positive $(U,V) \neq (1,1)$ and has contact order exactly $2r$. If $2r > E(\mathcal{N})$, its top U-face is $A_{\mathcal{M}}(V)$; if $2r > E(\mathcal{M})$, its top V-face is $A_{\mathcal{N}}(U)$.*

Proof. Both cyclotomic products are positive by theorem 8.1; the even powers are nonnegative and cannot vanish simultaneously away from $(1,1)$. Since $A_{\mathcal{M}}(1), A_{\mathcal{N}}(1) > 0$, the first nonzero homogeneous part has degree $2r$. The face statements follow by comparing degrees. □

Combining this proposition with theorems 7.1 and 8.2 gives the following exact audit.

Corollary 8.4 (Order-cover necessity). *Assume $2r > E(\mathcal{N})$. Let $p \mid S_1$, $p \nmid R_2S_2$, and $p \nmid \prod_{m\in\mathcal{M}} m$. For the first p-pole layer of the U-direction to cancel in (56), it is necessary that*

$$\mathrm{ord}_p(z_2 \bmod p) \in \mathcal{M}. \tag{57}$$

The symmetric condition holds for denominator primes of z_2.

Proof. The top U-face is $A_{\mathcal{M}}(V)$. First-layer cancellation requires it to vanish modulo p by theorem 7.2; theorem 8.2 translates this to (57). □

8.1 Depth, not merely residue order

Covering a prime modulo p cancels at most the first pole layer. If $v_p(S_1) = s$ is large, the full condition is

$$v_p(S_2^{E(\mathcal{M})} A_{\mathcal{M}}(z_2)) \geq s. \tag{58}$$

For a single factor with $p \nmid m$, the left side is $v_p(z_2^m - 1)$ once the exact order is m. Lifting-the-exponent arguments and effective p-adic logarithm bounds then show that a fixed order factor usually supplies only logarithmic depth in the exponent-vector height, while s can grow linearly. This is where recent sharp bounds for two p-adic logarithms can be inserted.

Research target 8.5 (Finite-pair low-cost deep order cover). For a family of finite or nonadjacent output near-unit pairs admitting a private denominator prime, let H be the maximum exponent-vector height. Determine whether the private denominator mass of the first coordinate can be covered to full p-adic depth by a product A_H evaluated at the second coordinate with

$$\deg A_H = o(H), \qquad \log \|A_H\|_1 = o(H), \tag{59}$$

and analogously in the reverse direction. Monomial factors are to be removed first, so the target concerns genuine unit-root capture.

The target is intentionally local and candidate-specific. It can matter for finite or nonadjacent configurations, but theorem 9.1 below shows that sufficiently deep adjacent pairs have no private structural denominator primes. The asymptotic endgame is therefore the shared-pole problem developed in section 9.

8.2 Why the order-cover formulation is useful

It separates four effects that are easily conflated.

1. **Toric aliases.** These vanish identically and are removed before any contact is counted.
2. **Monomial valuation transport.** This is a linear cone problem on the fixed valuation vectors of M, N and belongs to the tropical regime already studied.
3. **Finite private-prime unit-root capture.** This is the order-cover mechanism visible in the top faces of the cyclotomic prototype.
4. **Shared structural-face cancellation.** For sufficiently deep adjacent pairs, the same structural primes divide both denominators; multi-term initial forms must then cancel repeatedly on the shared Newton fan.

The decomposition is exact. The third mechanism remains a useful finite diagnostic, while the stable-support theorem below shows that the fourth is the asymptotic endgame.

Status: [proved here] for positivity, coefficient budget, residue-order detection, and the resulting necessary order-cover condition. Theorem 8.5 is an open finite-pair prototype. It is superseded asymptotically by the shared structural-face target of theorem 9.7.

9 Stable structural support and the shared-pole Newton fan

The private-prime analysis above is a useful local model, but it is not the asymptotic geometry of the actual adjacent continued-fraction near-units. For those pairs the numerator and denominator supports eventually stabilize. Thus the two coordinates do not exchange fresh denominator primes; they carry the same fixed structural primes at different linear depths. This changes the surviving endgame from private-prime order covering to simultaneous cancellation on shared p-adic Newton faces.

Keep the convergent notation

$$\nu_k = (-p_k, q_k), \qquad z_k = \frac{N^{q_k}}{M^{p_k}} = \frac{R_k}{S_k} \quad \text{in lowest terms.}$$

For a prime $\ell \mid MN$, put

$$m_\ell = v_\ell(M), \qquad n_\ell = v_\ell(N), \qquad u_{\ell,k} = v_\ell(z_k) = q_k n_\ell - p_k m_\ell. \tag{60}$$

Since $p_k/q_k \to \vartheta$, one has the exact decomposition

$$u_{\ell,k} = q_k(n_\ell - \vartheta m_\ell) + m_\ell(q_k\vartheta - p_k). \tag{61}$$

Theorem 9.1 (Stable structural support)**.** *Define*

$$\mathcal{P}_+ = \{\ell \mid MN : n_\ell - \vartheta m_\ell > 0\}, \qquad \mathcal{P}_- = \{\ell \mid MN : n_\ell - \vartheta m_\ell < 0\}. \tag{62}$$

Then $\mathcal{P}_+$ and $\mathcal{P}_-$ are disjoint and nonempty, and their union is the set of primes dividing MN. For all sufficiently large k,

$$R_k = \prod_{\ell \in \mathcal{P}_+} \ell^{u_{\ell,k}}, \qquad S_k = \prod_{\ell \in \mathcal{P}_-} \ell^{-u_{\ell,k}}. \tag{63}$$

Moreover, for every $\ell \mid MN$,

$$|u_{\ell,k}| = q_k |n_\ell - \vartheta m_\ell| + O_{M,N}(q_{k+1}^{-1}), \tag{64}$$

so each nonzero numerator or denominator depth is linear in q_k. In particular, for all sufficiently large adjacent pairs (z_k, z_{k+1}) there is no prime private to one reduced denominator: both denominators are supported on the same fixed set $\mathcal{P}_-$.

Proof. If $\ell \mid MN$, then $(m_\ell, n_\ell) \neq (0,0)$. The coefficient $n_\ell - \vartheta m_\ell$ cannot vanish, because otherwise either $m_\ell = 0 = n_\ell$ or $\vartheta = n_\ell/m_\ell \in \mathbb{Q}$, contradicting the irrationality of $\log 3/\log 2$. The convergent estimate $|q_k\vartheta - p_k| < q_{k+1}^{-1}$ in (61) therefore fixes the sign for all sufficiently large k and gives (64). Reduced numerator and denominator valuations are the positive and negative parts of $u_{\ell,k}$, proving (63).

Finally,

$$\sum_{\ell \mid MN} (n_\ell - \vartheta m_\ell) \log \ell = \log N - \vartheta \log M = 0.$$

Every summand coefficient is nonzero. Hence they cannot all have the same sign, so both $\mathcal{P}_+$ and $\mathcal{P}_-$ are nonempty. □

The theorem is easy to miss because z_k itself tends to 1 and alternates around it. The near-cancellation is not caused by small prime valuations. It is a cancellation between two fixed, linearly growing valuation profiles whose logarithmic weights almost balance.

9.1 Adjacent valuation transport

Let

$$\varepsilon_k = p_{k+1} q_k - p_k q_{k+1} \in \{\pm 1\}. \tag{65}$$

The unimodularity of adjacent convergents gives exact transport identities at every structural prime.

Proposition 9.2 (Adjacent valuation transport and toric balance)**.** *For every prime $\ell \mid MN$,*

$$q_{k+1} u_{\ell,k} - q_k u_{\ell,k+1} = \varepsilon_k m_\ell, \tag{66}$$

$$p_{k+1} u_{\ell,k} - p_k u_{\ell,k+1} = \varepsilon_k n_\ell. \tag{67}$$

Equivalently, in $\mathbb{Q}_{>0}^\times$,

$$\frac{z_k^{q_{k+1}}}{z_{k+1}^{q_k}} = M^{\varepsilon_k}, \tag{68}$$

$$\frac{z_k^{p_{k+1}}}{z_{k+1}^{p_k}} = N^{\varepsilon_k}. \tag{69}$$

Proof. Insert $u_{\ell,j} = q_j n_\ell - p_j m_\ell$ and use $p_{k+1}q_k - p_k q_{k+1} = \varepsilon_k$. The multiplicative identities follow by the same exponent calculation before taking valuations. $\square$

Thus the two shared pole depths are not independent random data. Their large linear parts are tied by a determinant-one defect equal to one fixed valuation of M or N. This exact coupling should be incorporated into any search rather than treated as an accidental correlation.

9.2 The shared-pole Newton polygon

Fix a sufficiently large adjacent pair and abbreviate

$$z_1 = z_k = R_1/S_1, \qquad z_2 = z_{k+1} = R_2/S_2.$$

Let

$$F(U,V) = \sum_{\substack{0\le i\le d\\ 0\le j\le e}} c_{ij}U^iV^j \in \mathbb{Z}[U,V], \qquad J_F = S_1^d S_2^e F(z_1,z_2) \in \mathbb{Z}. \tag{70}$$

For $\ell \in \mathcal{P}_-$ write $s_{\ell,j} = -u_{\ell,j} = v_\ell(S_j)$ for every sufficiently large j. For the fixed adjacent pair abbreviate

$$s_{\ell,1} = s_{\ell,k} = v_\ell(S_1), \qquad s_{\ell,2} = s_{\ell,k+1} = v_\ell(S_2). \tag{71}$$

Both depths are positive and of order q_k.

Theorem 9.3 (Exact shared-pole Newton law)**.** *For $\ell \in \mathcal{P}_-$ define*

$$\lambda_{ij}^{(\ell)} = v_\ell(c_{ij}) + (d-i)s_{\ell,1} + (e-j)s_{\ell,2} \tag{72}$$

for $c_{ij} \neq 0$, and let $\lambda_\ell = \min \lambda_{ij}^{(\ell)}$. Then

$$v_\ell(J_F) \ge \lambda_\ell. \tag{73}$$

If the minimum is attained at a unique exponent pair, equality holds. More generally, write $S_t = \ell^{s_{\ell,t}} S'_t$ with $\ell \nmid S'_t$. After division by ℓ^{λ_ℓ}, the residue of J_F modulo ℓ is the sum of the residues of the terms on the minimizing face:

$$\mathrm{in}_\ell(F; z_1, z_2) = \sum_{\lambda_{ij}^{(\ell)} = \lambda_\ell} \overline{\ell^{-v_\ell(c_{ij})}c_{ij}}\,\overline{R_1}^{\,i}\,\overline{R_2}^{\,j}\,\overline{S'_1}^{\,d-i}\,\overline{S'_2}^{\,e-j} \in \mathbb{F}_\ell. \tag{74}$$

Consequently

$$v_\ell(J_F) > \lambda_\ell \iff \mathrm{in}_\ell(F; z_1, z_2) = 0. \tag{75}$$

Iterating after division by the current power of ℓ gives an exact finite-depth cancellation cascade.

Proof. Expand

$$J_F = \sum_{i,j} c_{ij} R_1^i R_2^j S_1^{d-i} S_2^{e-j}.$$

Because z_1, z_2 are reduced and ℓ divides both denominators, $\ell \nmid R_1R_2S'_1S'_2$. The valuation of each summand is exactly (72). The ultrametric inequality proves (73); a unique minimum cannot cancel. Factoring the common power ℓ^{λ_ℓ} gives (74) and (75). $\square$

Corollary 9.4 (Unit-corner survival)**.** *If c_{de} is an ℓ-adic unit, then (d,e) is the unique minimizing exponent pair, $v_\ell(J_F)=0$, and the full pole*

$$\ell^{ds_{\ell,1}+es_{\ell,2}} \tag{76}$$

survives in the reduced denominator of $F(z_1,z_2)$.

This is the shared-prime analogue of the earlier face-monic obstruction. It also shows why removing the top-right monomial is only the beginning: once the minimizing set has more than one term, one must force its initial form to vanish at every stable denominator prime, then repeat the argument at every higher ℓ-adic layer.

Example 9.5 (The positive two-axis mask)**.** For

$$F_r(U,V)=(U-1)^{2r}+(V-1)^{2r}$$

one has

$$J_{F_r}=G_1^{2r}S_2^{2r}+G_2^{2r}S_1^{2r}. \tag{77}$$

At $\ell\in\mathcal{P}_-$ the two displayed terms have valuations $2rs_{\ell,2}$ and $2rs_{\ell,1}$. If the depths are unequal, the smaller one is unique and

$$v_\ell(J_{F_r})=2r\min(s_{\ell,1},s_{\ell,2}), \tag{78}$$

so the reduced denominator retains $\ell^{2r\max(s_{\ell,1},s_{\ell,2})}$. If the depths tie, cancellation is possible only through the explicit two-term residue in (77). Positivity at the real place therefore does not by itself buy denominator compression.

Lemma 9.6 (Two-term contact cannot vanish at the output)**.** *Let z_1,z_2 be the adjacent output near-units. A nonzero Laurent polynomial with at most two monomials cannot satisfy both $F(1,1)=0$ and $F(z_1,z_2)=0$.*

Proof. The condition at $(1,1)$ makes the two coefficients opposite, so after multiplying by a monomial one has $F=c(U^aV^b-1)$ with $(a,b)\neq(0,0)$. Vanishing at (z_1,z_2) would give $z_1^az_2^b=1$. Adjacent relation vectors form a unimodular basis of $\mathbb{Z}^2$, so z_1,z_2 generate the same rank-two group as M,N and are multiplicatively independent. Thus $a=b=0$, a contradiction. □

The lemma does not forbid a two-term initial form from canceling modulo one prime. It says that no single binomial supplies exact cancellation at every depth and every place. A viable mask must use genuinely multi-term, prime-dependent Newton faces.

Research target 9.7 (Shared structural-face capture)**.** Construct, for some sufficiently large adjacent pair, a nonzero mask F_k with contact r_k such that:

1. its shifted coefficient norm is subheight and its contact supplies the archimedean gain required by the scalar criterion of section 11;
2. for every $\ell\in\mathcal{P}_-$, the successive initial forms in (74) vanish deeply enough that

$$\sum_{\ell\in\mathcal{P}_-}\left(d_ks_{\ell,k}+e_ks_{\ell,k+1}-v_\ell(J_{F_k})\right)_+\log\ell=O_x((d_k+e_k)\log q_k); \tag{79}$$

3. the mask is certified nonzero at (z_k,z_{k+1}) and is non-toric.

For a fixed hypothetical candidate, one mask satisfying these finite conditions and the strict scalar inequality contradicts that candidate. A construction that works for every hypothetical candidate would prove the Alaoglu–Erdős conjecture.

Status: **[proved here]** for stable support, adjacent valuation transport, the shared-pole Newton law, unit-corner survival, the positive-mask calibration, and the two-term nonvanishing lemma. Theorem 9.7 is the new open endgame. The earlier private-prime and cyclotomic sections remain valid finite diagnostics, but theorem 9.1 shows that they are not the asymptotic adjacent-convergent mechanism.

10 Three quantitative barriers for naive mask constructions

The preceding sections identify the exact arithmetic objects. We now combine them with the continued-fraction scale. The conclusions are negative, but they are useful negative results: they rule out three broad architectures while leaving one sharply defined escape route.

Let p_n/q_n be the continued-fraction convergents to ϑ, and put

$$\nu_n = (-p_n, q_n), \qquad \delta_n = q_n \log 3 - p_n \log 2, \qquad z_n = \sigma_{\nu_n} = \frac{N^{q_n}}{M^{p_n}} = e^{x\delta_n}. \tag{80}$$

Write $z_n = R_n/S_n$ in lowest terms and set

$$\Delta_n = \max\{|x\delta_n|, |x\delta_{n+1}|\}. \tag{81}$$

For all sufficiently large n, $\Delta_n < 1/4$.

We use one standard external input. An effective lower bound for linear forms in $\log 2, \log 3$ gives constants $C > 0$ and $\kappa > 1$ such that

$$|q\vartheta - p| \geq Cq^{-\kappa} \qquad (p \in \mathbb{Z},\ q \geq 1). \tag{82}$$

For convergents this implies $q_{n+1} \ll q_n^{\kappa}$ and hence

$$\log \frac{1}{\Delta_n} = O_x(\log q_n). \tag{83}$$

Only the existence of some effective exponent is needed; no numerical optimization enters.

10.1 The full-clearing Taylor envelope cannot become subunit

Let $F_n \in \mathbb{Z}[U, V]$ be nonzero, of bidegree at most (d_n, e_n) and contact order r_n at $(1, 1)$. Write

$$\widetilde{F}_n(X, Y) = F_n(1 + X, 1 + Y).$$

Clearing the visible denominators and applying theorem 4.4 gives

$$|J_n| \leq S_n^{d_n} S_{n+1}^{e_n} \left\|\widetilde{F}_n\right\|_1 (2\Delta_n)^{r_n}, \qquad J_n := S_n^{d_n} S_{n+1}^{e_n} F_n(z_n, z_{n+1}) \in \mathbb{Z}. \tag{84}$$

Theorem 10.1 (Full-clearing envelope barrier)**.** *Assume $d_n + e_n > 0$. Then the right side of (84) is greater than 1 for all sufficiently large n, uniformly over all choices of F_n. Consequently the ordinary argument*

$$\textit{clear every visible denominator} + \textit{bound by Taylor contact}$$

can never prove $J_n = 0$ or produce a nonzero integer of absolute value below one.

Proof. By theorem 2.3, norm equivalence, and $p_n \asymp q_n$, there is a constant $c_x > 0$ such that

$$\log S_n \geq c_x q_n - O_x(1), \qquad \log S_{n+1} \geq c_x q_{n+1} - O_x(1)$$

for all large n. The contact ceiling theorem 4.3 gives $r_n \leq d_n + e_n$, while $\left\|\widetilde{F}_n\right\|_1 \geq 1$. Therefore the logarithm of the right side in (84) is at least

$$c_x q_n(d_n + e_n) - O_x(d_n + e_n) - (d_n + e_n) O_x(\log q_n),$$

where (83) was used in the last term. This is positive for all sufficiently large n. □

Remark 10.2 (What the theorem does not say)*.* The theorem does not lower-bound the actual value $|J_n|$ by the displayed envelope. It says that this particular triangle-inequality proof cannot make the integer subunit. The two possible escapes are cancellation inside the archimedean sum, or cancellation in the *reduced* denominator before clearing. The first is an auxiliary-function problem; the second is the shared-pole Newton-face problem isolated in section 9.

10.2 Imported gap transport and the new shared-gap consequence

For a relation vector $\nu = (-p, q)$ put

$$z_\nu = \frac{N^q}{M^p} = \frac{R_\nu}{S_\nu}, \qquad G_\nu = R_\nu - S_\nu,$$

where the fraction is reduced. The supplied report already proves the determinant-transport law for pairs of binomial gaps. Before importing it, there is a useful reduced-gap observation that removes the usual exceptional support.

Lemma 10.3 (Structural primes cannot be shared by independent reduced gaps)**.** *Let $\nu, \omega \in \mathbb{Z}^2$ be linearly independent. Then*

$$\gcd(G_\nu, G_\omega) \textit{ is coprime to } MN. \tag{85}$$

Proof. Suppose that a prime $\ell \mid MN$ divides G_ν. Since $\gcd(R_\nu, S_\nu) = 1$, a prime dividing $R_\nu - S_\nu$ divides neither R_ν nor S_ν. Hence

$$v_\ell(z_\nu) = 0, \qquad \nu \cdot (v_\ell(M), v_\ell(N)) = 0.$$

If the same prime divides G_ω, then ω is orthogonal to the same nonzero valuation vector. Its kernel in $\mathbb{Q}^2$ is one-dimensional, forcing ν and ω to be linearly dependent, a contradiction. □

Lemma 10.4 (Inherited determinant transport for reduced gaps)**.** *Let $p, q, r, s \in \mathbb{N}$ and let $\nu = (-p, q)$ and $\omega = (-r, s)$ be independent, with $D = |ps - rq|$. Then*

$$\gcd(G_\nu, G_\omega) \mid \gcd(M^D - 1, N^D - 1). \tag{86}$$

Proof. This is the gap-determinant theorem proved in the supplied report, specialized to reduced output gaps. For completeness of the specialization, a prime power dividing both reduced gaps is coprime to MN by theorem 10.3; it therefore divides the two unreduced gaps $N^q - M^p$ and $N^s - M^r$. The inherited determinant transport gives $M^D \equiv N^D \equiv 1$ modulo that prime power, proving (86) prime by prime. ☐

For adjacent convergents, $D = 1$. Thus

$$\gcd(G_n, G_{n+1}) \mid \gcd(M-1, N-1), \tag{87}$$

with no exceptional structural factor. Combined with the new gap-ideal divisor (31), this contributes at most $\exp(O_x(r_n))$, whereas the visible denominator scale is $\exp(\Omega_x(q_n(d_n+e_n)))$. Since $r_n \le d_n + e_n$, adjacent shared-gap amplification is exponentially too small.

The same mask-level conclusion survives nonadjacent pairs.

Corollary 10.5 (Shared-gap mask barrier)**.** *Consider independent pairs $\nu_j, \omega_j \in \mathbb{Z}^2$ with*

$$H_j = \|\nu_j\|_1 + \|\omega_j\|_1 \longrightarrow \infty, \qquad \max(|\delta_{\nu_j}|, |\delta_{\omega_j}|) \longrightarrow 0.$$

Let $g_j = \gcd(G_{\nu_j}, G_{\omega_j})$ and let $r_j \ge 1$ be any mask contact order. Then

$$\log g_j = o(H_j), \qquad \log(g_j^{r_j}) = o(H_j r_j). \tag{88}$$

Consequently the universal shared-gap divisor supplied by contact cannot pay a denominator tax of order $H_j(d_j+e_j)$ when $r_j \le d_j + e_j$.

Proof. Put $D_j = |\det(\nu_j, \omega_j)|$. The exact identities (10) give $D_j \ll H_j \max(|\delta_{\nu_j}|, |\delta_{\omega_j}|) = o(H_j)$. By theorem 10.4, $g_j \mid \gcd(M^{D_j} - 1, N^{D_j} - 1)$. If D_j is bounded, then g_j is bounded. If $D_j \to \infty$, the theorem of Bugeaud–Corvaja–Zannier gives

$$\log \gcd(M^{D_j} - 1, N^{D_j} - 1) = o(D_j),$$

because M, N are multiplicatively independent. Thus $\log g_j = o(H_j)$ in either case; multiplication by r_j proves the second estimate. ☐

10.3 The private-prime obstruction is local; the shared-pole obstruction is asymptotic

For finite or nonadjacent pairs that possess a private denominator prime, the face-monic obstruction of theorem 7.3 remains exact. It eliminates powers of one-variable gaps, masks with unit or monomial top faces, and generic sparse masks lacking explicit opposite-coordinate order control.

For sufficiently deep adjacent convergents, however, theorem 9.1 shows that there are no private structural denominator primes at all. The obstruction changes form: at every $\ell \in \mathcal{P}_-$ both coordinates have poles of linear depth, and the minimizing terms are selected by the shared weight

$$(i,j) \longmapsto v_\ell(c_{ij}) + (d-i)s_{\ell,k} + (e-j)s_{\ell,k+1}.$$

A unit top-right corner leaves the whole pole, while a missing corner creates a slanted Newton face whose initial form must vanish. Near-total compression requires this vanishing to persist through almost the complete depth at every prime in the fixed set $\mathcal{P}_-$. Thus the three quantitative barriers leave one architecture: multi-term, prime-specific cancellation on shared Newton faces, coupled by the exact transport identities (66)–(67).

Status: [proved here] for the full-clearing envelope barrier, structural-prime lemma, and shared-gap mask corollary; the determinant-transport lemma is inherited from the supplied report; [external theorem] for the effective two-logarithm estimate and the Bugeaud–Corvaja–Zannier gcd bound. The conclusions concern the specified proof architectures, not the truth of the Alaoglu–Erdős conjecture.

11 The reduced-denominator endgame

The preceding no-go theorem used the visible denominator $S_n^{d_n} S_{n+1}^{e_n}$. That is deliberately crude. The actual arithmetic quantity is the reduced denominator of the rational mask value.

Let $F \in \mathbb{Z}[U, V]$ have contact order r, and suppose

$$F(z_1, z_2) = \frac{a_F}{b_F} \quad \text{in lowest terms}, \qquad b_F > 0.$$

Set $\eta_i = \log z_i$ and $\Delta = \max(|\eta_1|, |\eta_2|)$.

Criterion 11.1 (Exact scalar mask criterion)**.** *Assume $0 < \Delta \le 1/4$ and $F(z_1, z_2) \ne 0$. If*

$$r \log \frac{1}{2\Delta} > \log b_F + \log \|F(1 + X, 1 + Y)\|_1\,, \tag{89}$$

then no hypothetical counterexample producing z_1, z_2 can exist.

Proof. A nonzero reduced rational number satisfies $|F(z_1, z_2)| \ge 1/b_F$. On the other hand, theorem 4.4 gives

$$|F(z_1, z_2)| \le \|F(1 + X, 1 + Y)\|_1\,(2\Delta)^r.$$

Condition (89) makes the latter strictly smaller than $1/b_F$. □

This elementary criterion is the precise endgame. It neither assumes nor hides a determinant. Every candidate construction can be audited by three numbers: contact r, shifted coefficient norm, and the *actual reduced denominator* b_F.

11.1 The denominator-compression index

If F has bidegree at most (d, e), then b_F divides the visible denominator $B_F = S_1^d S_2^e$. Define the integer compression factor and normalized compression index

$$K_F = \frac{B_F}{b_F}, \qquad \mathfrak{c}(F; z_1, z_2) = \frac{\log K_F}{d \log S_1 + e \log S_2}, \tag{90}$$

with the convention $\mathfrak{c} = 0$ when $B_F = 1$. Thus $0 \le \mathfrak{c} \le 1$. Criterion (89) becomes

$$\log K_F > d \log S_1 + e \log S_2 + \log \|F(1 + X, 1 + Y)\|_1 - r \log \frac{1}{2\Delta}. \tag{91}$$

Theorem 11.2 (Near-total compression is necessary on adjacent convergents)**.** *Let F_n be masks on (z_n, z_{n+1}) with bidegrees (d_n, e_n), contact r_n, and $F_n(z_n, z_{n+1}) \neq 0$. Assume*

$$\log \|F_n(1+X, 1+Y)\|_1 = o\big(q_n(d_n + e_n)\big). \tag{92}$$

If the scalar criterion succeeds for infinitely many n, then

$$\mathfrak{c}(F_n; z_n, z_{n+1}) = 1 - o(1) \tag{93}$$

along that subsequence. More quantitatively, the uncanceled denominator obeys

$$\log b_{F_n} \leq O_x\big((d_n + e_n)\log q_n\big) - \log \|F_n(1+X, 1+Y)\|_1 . \tag{94}$$

Proof. The contact ceiling gives $r_n \leq d_n + e_n$, and (83) gives $r_n \log(1/(2\Delta_n)) = O_x((d_n + e_n)\log q_n)$. Criterion (89) then implies (94). By theorem 2.3, the visible denominator has logarithm $\Omega_x(q_n(d_n + e_n))$. Under (92), subtracting (94) from $\log B_{F_n}$ gives $\log K_{F_n} = (1 - o(1))\log B_{F_n}$. □

Thus "some cancellation" is not enough. The mask must remove all but a polynomial-height fraction of an exponentially large visible denominator. Stable structural support and the shared-pole Newton law make that requirement exact prime by prime.

11.2 Exact primewise reduced-denominator formula

For all sufficiently large n, every prime divisor of $S_n S_{n+1}$ lies in the fixed set $\mathcal{P}_-$. Put

$$J_{F_n} = S_n^{d_n} S_{n+1}^{e_n} F_n(z_n, z_{n+1})$$

and, for $\ell \in \mathcal{P}_-$, define the uncanceled pole depth

$$\rho_{\ell,n}(F_n) = \big(d_n s_{\ell,n} + e_n s_{\ell,n+1} - v_\ell(J_{F_n})\big)_+. \tag{95}$$

Proposition 11.3 (Exact shared-prime denominator law)**.** *If $F_n(z_n, z_{n+1}) \neq 0$ and b_{F_n} is its reduced denominator, then*

$$b_{F_n} = \prod_{\ell \in \mathcal{P}_-} \ell^{\rho_{\ell,n}(F_n)}, \qquad \log b_{F_n} = \sum_{\ell \in \mathcal{P}_-} \rho_{\ell,n}(F_n) \log \ell. \tag{96}$$

Proof. The unreduced rational value is $J_{F_n}/(S_n^{d_n} S_{n+1}^{e_n})$. At a prime $\ell \in \mathcal{P}_-$ the denominator has valuation $d_n s_{\ell,n} + e_n s_{\ell,n+1}$; reduction against the numerator J_{F_n} leaves exactly (95). No other prime occurs in the visible denominator. □

Corollary 11.4 (Scalar success forces simultaneous shared-face capture)**.** *Assume the subheight coefficient condition (92). If the scalar criterion succeeds along infinitely many adjacent pairs, then*

$$\sum_{\ell \in \mathcal{P}_-} \rho_{\ell,n}(F_n) \log \ell = O_x\big((d_n + e_n)\log q_n\big) \tag{97}$$

along that subsequence. Since $\mathcal{P}_-$ is fixed and every visible pole depth is $\Theta_x(q_n)$, the mask must capture a $1 - o(1)$ fraction of the pole depth at every stable denominator prime that contributes a positive proportion of the visible height.

Proof. Combine theorem 11.3 with (94). The final assertion follows from (64) and the finiteness of $\mathcal{P}_-$. A primewise residual may be redistributed among the finitely many primes, but the weighted total is only polylogarithmic while each uncanceled linear fraction would contribute $\Theta(q_n)$. □

This is the precise surviving endgame. At each stable denominator prime, the Newton initial form (74) must cancel; after division by the first canceled layer, the next initial form must cancel; and so on through almost the complete linear depth. The exact transport laws show that the two depth vectors differ by a determinant-one defect, so the required cancellations are strongly coupled across adjacent indices.

11.3 Constructive and obstructive targets

The constructive target is theorem 9.7. Cyclotomic faces remain a useful finite source of residue identities, but stable support means that a successful mask must combine them into multi-term shared faces rather than assign a private prime to the opposite coordinate. Positivity, a Sturm certificate, or another exact argument must also keep the final rational value nonzero.

The complementary obstruction target is to prove that every nonzero, non-toric mask of contact r and subheight shifted coefficient norm leaves a positive proportion of the shared pole mass uncanceled:

$$\log b_F \geq c_x(d \log S_1 + e \log S_2) - O_x(r \log H) \tag{98}$$

for some $c_x > 0$, where H is the exponent-vector height. Even a fixed positive c_x would contradict the necessary polyheight bound (94). By theorem 9.3, this is a finite-prime theorem about repeated cancellation on shared Newton faces, not an archimedean approximation theorem.

Status: **[proved here]** for the scalar criterion, compression reformulation, near-total-compression theorem, exact shared-prime denominator law, and forced simultaneous capture. Theorem 9.7 and (98) are the constructive and obstructive open endgames.

---

12 Exact and numerical reconnaissance

The computations in this section are diagnostic. None is used as an unproved premise in the theorems above. Their purpose is to test constants, catch false high-contact masks that are actually toric, and identify the scale a constructive search must beat.

12.1 Continued-fraction near-unit bases

The first convergents to $\vartheta = \log 3/\log 2$ and their logarithmic errors are shown below. The last column is

$$\mathcal{U}_n = \frac{2H_n \max(|\delta_{n-1}|, |\delta_n|)}{\log 3}, \qquad H_n = \max(p_{n-1}, q_{n-1}, p_n, q_n). \tag{99}$$

The determinant-one identity forces $\mathcal{U}_n \geq 1$. Values close to one indicate that the simple uncertainty bound is already near the observed scale.

n	p_n/q_n	$\delta_n = q_n \log 3 - p_n \log 2$	$q_n\lvert\delta_n\rvert$	$\mathcal{U}_n$
0	1/1	$4.0546510811\times 10^{-1}$	4.0547×10^{-1}	–
1	2/1	$-2.8768207245\times 10^{-1}$	2.8768×10^{-1}	1.4763
2	3/2	$1.1778303566\times 10^{-1}$	2.3557×10^{-1}	1.5711
3	8/5	$-5.2116001139\times 10^{-2}$	2.6058×10^{-1}	1.7155
4	19/12	$1.3551033379\times 10^{-2}$	1.6261×10^{-1}	1.8016
5	65/41	$-1.1462901003\times 10^{-2}$	4.6998×10^{-1}	1.6037
6	84/53	$2.0881323758\times 10^{-3}$	1.1067×10^{-1}	1.7532
7	485/306	$-1.0222391238\times 10^{-3}$	3.1281×10^{-1}	1.8437
8	1054/665	$4.3654128136\times 10^{-5}$	2.9030×10^{-2}	1.9615
9	24727/15601	$-1.8194176704\times 10^{-5}$	2.8385×10^{-1}	1.9657
10	50508/31867	$7.2657747287\times 10^{-6}$	2.3153×10^{-1}	1.6355
11	125743/79335	$-3.6626272464\times 10^{-6}$	2.9059×10^{-1}	1.6630
12	176251/111202	$3.6031474823\times 10^{-6}$	4.0068×10^{-1}	1.1559
13	301994/190537	$-5.9489764120\times 10^{-8}$	1.1334×10^{-2}	1.9807

The large partial quotient 55 following the thirteenth convergent creates a spectacularly small single error, but the adjacent pair still obeys the determinant-one floor. This is the concrete form of the geometry-of-numbers obstruction: a single exceptional direction does not create two independent exceptional directions.

12.2 Symbolic mask checks

The univariate Thue–Morse masks

$$P_r(T) = \prod_{j=0}^{r-1}(1 - T^{2^j})$$

have the following exact data.

r	degree	support	coefficient ℓ^1 norm	multiplicity at 1
1	1	2	2	1
2	3	4	4	2
3	7	8	8	3
4	15	16	16	4
5	31	32	32	5
6	63	64	64	6
7	127	128	128	7

The growth is attractive from a moment-cancellation viewpoint, but theorem 5.1 shows that the cleared value is still a product of integer gap factors. The computational table therefore confirms, rather than evades, the one-direction obstruction.

For comparison, the positive transverse masks

$$F_r(U, V) = (U - 1)^{2r} + (V - 1)^{2r} \tag{100}$$

have exact contact $2r$, bidegree $(2r, 2r)$, and support $4r + 1$. Their ordinary coefficient ℓ^1 norm is 2^{2r+1}, while the shifted mask is simply $X^{2r} + Y^{2r}$ and therefore has shifted norm 2. The first values are

r	contact	support	ordinary ℓ^1	shifted ℓ^1
1	2	5	8	2
2	4	9	32	2
3	6	13	128	2
4	8	17	512	2

They are nonzero by positivity and have an exceptionally small shifted norm. At a shared denominator prime, however, theorem 9.5 shows that unequal pole depths leave the larger depth completely visible, while equal depths merely create one explicit two-term residue test. This is a compact calibration of the main tradeoff: nonvanishing and contact are easy; repeated shared-prime denominator compression is the hard resource.

The cyclotomic-faced examples

$$(V+1)(U-1)^{2r}+(U+1)(V-1)^{2r}$$

were checked symbolically for $1 \le r \le 4$. Their contact is exactly $2r$, and their top faces are $V+1$ and $U+1$ once $2r > 1$. They are among the smallest explicit positive examples in which a finite private prime can be canceled by a genuine order condition on the opposite coordinate. Stable support shows that an asymptotic construction must combine such factors into tied multi-term faces at the same structural primes, rather than assign fresh primes between coordinates.

12.3 Reproducible source

The following compact script reproduces the continued-fraction and uncertainty table. It uses high precision for reconnaissance; a proof-producing implementation would replace the floating comparisons by directed logarithm intervals.

Listing 1: Continued-fraction and uncertainty reconnaissance.

```
import mpmath as mp

mp.mp.dps = 100
log2, log3 = mp.log(2), mp.log(3)
theta = log3 / log2

# Continued fraction and convergents.
a = []
y = theta
for _ in range(18):
    z = int(mp.floor(y))
    a.append(z)
    y = 1 / (y - z)

p_m2, p_m1 = 0, 1
q_m2, q_m1 = 1, 0
conv = []
for z in a:
    p = z * p_m1 + p_m2
    q = z * q_m1 + q_m2
    delta = q * log3 - p * log2
    conv.append((p, q, delta))
    p_m2, p_m1 = p_m1, p
    q_m2, q_m1 = q_m1, q

for n, (p, q, delta) in enumerate(conv):
    if n == 0:
        ratio = None
```

```
    else:
        pp, qq, dd = conv[n - 1]
        H = max(p, q, pp, qq)
        ratio = 2 * H * max(abs(delta), abs(dd)) / log3
    print(n, p, q, delta, q * abs(delta), ratio)
```

For symbolic contact checks, one may expand $F(1+X, 1+Y)$ and find the least total degree of a nonzero monomial. Toric masks must first be reduced modulo I_ν; otherwise an identically zero lattice relation can masquerade as arbitrarily high contact.

Status: [exact computation] for the symbolic polynomial identities; numerical continued-fraction values are reconnaissance only. No theorem in the report depends on the printed decimals.

13 Lean-oriented formalization plan

The new framework is unusually suitable for formalization because its core is finite algebra, integer valuation bookkeeping, and continued-fraction determinant identities. No Four Exponentials input is needed to kernel-check the exact reductions. A natural module split is as follows.

Proposed module	Main statements
`RelationLattice`	Relation vectors, δ_ν, determinant identities (10), adjacent-convergent orientation, and the two-direction uncertainty bound.
`MonomialToricKernel`	Finite-support monomial map, aggregation by exponent vector, universal kernel equivalence, and invariance under unimodular basis changes.
`ContactMoments`	Shifted-polynomial contact, falling-factorial moments, ordinary moment equivalence, degree ceiling, support ceiling, and the local ℓ^1 estimate.
`UnivariateGapPower`	Divisibility of $S^d P(R/S)$ by $(R-S)^r$, including Laurent normalization and the Thue–Morse specialization.
`BivariateGapIdeal`	Cleared evaluation formula (29), membership in $(G_1, G_2)^r$, gcd-power divisibility, and the primewise tropical floor.
`PrivateLeadingFace`	The finite private-prime prototype: unique-minimum leading face, unit-face survival, and exact first-layer cancellation.
`CyclotomicFiniteFace`	Homogenized cyclotomic evaluation, finite-field order criterion, positivity of cross-faced masks, and finite-depth valuation conditions.
`StableSupport`	Equations (60)– (63), nonemptiness of $\mathcal{P}_\pm$, linear pole depths, and adjacent valuation transport (66)– (67).
`SharedPoleNewton`	Exact term weights (72), unique-minimum equality, initial-form cancellation, unit-corner survival, and the positive-mask calibration.
`ReducedGapTransport`	Lemma 10.4, structural-prime exclusion, and the shared-gap mask corollary.

Proposed module	Main statements
`ReducedDenominator`	Scalar criterion, compression factor, exact shared-prime denominator law (96), and the finite implication from a certified mask to contradiction.

A low-risk theorem order is:

1. formalize contact, gap-ideal algebra, and toric normal forms over an arbitrary commutative domain;
2. specialize to $\mathbb{Z}$, rational evaluations, and prime valuations;
3. formalize continued fractions, support stabilization, and the determinant-one transport laws;
4. add the shared-pole Newton fan and exact reduced-denominator formula;
5. only then connect the finite certificate to real powers and effective logarithmic recurrence.

The analytic interfaces can remain explicit hypotheses at first:

- $z_i = e^{\eta_i}$ and $|e^{\eta_i} - 1| \le 2|\eta_i|$ for $|\eta_i| \le 1/4$;
- effective recurrence $q_{n+1} \le C q_n^{\kappa}$;
- exact positivity or nonvanishing of the selected shared-face mask.

The first and third are routine real algebra/analysis once a concrete mask is given; the second can be imported through a named linear-forms interface. This separation keeps the trust boundary visible.

A proof-producing search can output a compact certificate containing:

1. the exponent support and integer coefficients of F;
2. exact moment equations certifying contact and a toric normal form proving the mask is not an identically vanishing lattice relation;
3. a factorization, positivity, or Sturm certificate proving nonvanishing;
4. exact reduced fractions $z_n = R_n/S_n$, $z_{n+1} = R_{n+1}/S_{n+1}$ and the stable sets $\mathcal{P}_\pm$;
5. the cleared integer J_F and its valuation at every prime in $\mathcal{P}_-$, including the successive shared-face initial-form cancellations;
6. the exact reduced denominator b_F from (96);
7. rational interval bounds verifying (89).

Every item except the final exponential estimate is finite integer arithmetic. The final estimate can use directed intervals for $\log 2$ and $\log 3$.

Status: [open target] only. No new Lean files are claimed in this report. The listed statements are designed as independent additions rather than modifications of the existing corpus.

14 Public-status audit and literature placement

The competitive framing in the prompt contains claims of very recent AI-assisted results. Because none of those claims is a mathematical premise for the present report, the correct standard is conservative: distinguish a public, inspectable result from an announcement, a provisional solution, or a private rumor.

14.1 What could be checked publicly on 22 August 2026

A short explicit three-dimensional counterexample to the Jacobian conjecture was publicly announced by Levent Alpöge with credit to Claude Fable, and independent expository accounts verify the displayed map directly; the two-dimensional case remains open [9, 10]. This is a genuine public mathematical development, but it has no known implication for the logarithmic period relation studied here.

Epoch AI's FrontierMath page for the twelve unresolved Hadamard orders below 2000 reports that Claude Fable 5.2 Pro submitted solutions for eleven orders and that a twelfth construction was found but not yet fully written up; the page explicitly labels the status provisional [11]. The corresponding Epoch page for elliptic curves of large rank still listed the problem as unsolved and recorded rank at least 29, not a publicly certified rank 30 construction, when accessed for this report [12]. Public posts and news reports may be ahead of that page, but without a curve, independent points, and a checkable rank certificate, the claimed rank-30 development is not used here.

No public paper, preprint, Lean repository, theorem statement, or inspectable proof describing the rumored new Alaoglu–Erdős breakthrough was located in the web audit. Absence of a search result is not evidence that no private work exists. It means only that there is no public technical input to import. Accordingly, every mathematical claim in this report is proved here, marked as an external theorem, or explicitly stated as a target.

14.2 How the new method sits relative to known work

The algebraic kernel in section 3 is part of the classical theory of lattice and binomial ideals [5, 6]. The new point is the use of that kernel as a *sanity filter* for continued-fraction near-unit masks: exact recurrence relations among many convergents are toric identities and hence vanish on both the input and output grids.

Prouhet–Tarry–Escott and Thue–Morse constructions are classical sources of moment cancellation [7]. The exact cleared-gap theorem theorem 5.1 identifies why every such one-direction construction is arithmetically inert here. The bivariate gap ideal, stable structural-support theorem, shared-pole Newton law, exact primewise denominator formula, and shared-face capture target are the genuinely problem-specific additions.

The effective recurrence estimate (82) belongs to the theory of linear forms in logarithms [2, 3]. The reduced-gap estimate in theorem 10.5 uses Bugeaud–Corvaja–Zannier [4]. These theorems control the size of available near relations and shared divisors; they do not address the bilinear logarithmic determinant directly.

The supplied unified report already explains the Four Exponentials barrier and develops interpolation determinants, Kummer theory, compact orbits, natural boundaries, moment methods, local rank, Smith profiles, and numerous other directions [13]. This supplement deliberately does not repeat those surveys. A literal source audit found no section organized around Prouhet or Thue–Morse masks, toric/lattice ideals, augmentation-order gap ideals, stable structural support, or the shared-pole Newton fan developed here. The present report is therefore an extension rather than a rearrangement of that manuscript.

Status: [external theorem] for the dated public-status statements and cited literature. None of the news claims is used as a proof input.

15 Conclusions and prioritized next steps

No unconditional proof of the Alaoglu–Erdős conjecture is claimed. The progress is a sharp reduction of a previously unseparated auxiliary-function architecture and, most importantly, an exact identification of the denominator mechanism that remains after the natural private-prime model fails.

15.1 What has been proved in this report

1. **Exact height of every exponent-lattice near-unit.** The reduced numerator and denominator are governed by the valuation norm $L_{M,N}$, so a good real relation still has exponential rational height.
2. **Toric collapse of many-convergent identities.** Universal monomial identities among any number of relation coordinates form a lattice ideal independent of the multiplicatively independent base pair. Exact continued-fraction recurrences therefore vanish on both the input and output grids and cannot create a nonzero small integer.
3. **Exact one-direction no-go theorem.** Contact of order r at 1 forces a factor $(T-1)^r$; after clearing denominators, the mask value contains the rth power of the corresponding integer output gap. This closes all one-direction Prouhet, Thue–Morse, and finite-difference masks at once.
4. **Multivariate augmentation-to-gap transfer.** Contact order r at $(1,1)$ places every cleared bivariate value in $(G_1, G_2)^r$, with an exact primewise tropical floor and a scalar product-formula criterion.
5. **Stable structural support.** For adjacent convergents, every prime dividing MN has an eventually fixed numerator or denominator sign. Both reduced denominators are supported on the same nonempty finite set $\mathcal{P}_-$, and every pole depth grows linearly with the convergent denominator. Thus fresh/private structural primes are not the asymptotic resource.
6. **Shared-pole Newton law.** At each $\ell \in \mathcal{P}_-$, every cleared monomial has the exact weight (72). A unique minimum survives; a tied face cancels one layer exactly when its explicit initial form vanishes. Near-total compression requires this cancellation to repeat through almost the full depth at every stable denominator prime.
7. **Exact reduced-denominator endgame.** Formula (96) expresses the denominator as the sum of the uncanceled shared pole depths. The scalar criterion forces their total weighted residual to be only $O((d+e)\log q)$, while the visible pole mass is $\Theta(q(d+e))$.

8. **Quantitative no-go results.** Full visible-denominator clearing cannot become subunit; shared reduced-gap divisibility is fixed-size for adjacent directions and subheight for nonadjacent directions; unit-corner and generic unique-face masks leave full pole layers.

15.2 The surviving research program

The highest-value next step is a structured search over shared Newton faces, not a broader search over arbitrary high-contact coefficients.

1. **Prime-coupled shared-face search.** For the finite set $\mathcal{P}_-$, impose the exact linear congruences of section C so that the cleared numerator captures all but $O(\log q)$ layers at every prime. Optimize the rigorous scalar margin

$$r \log \frac{1}{2\Delta} - \log b_F - \log \|F(1+X, 1+Y)\|_1 \, .$$

A positive interval-certified margin rules out the hypothetical candidate.

2. **Exploit determinant-one transport.** The depth vectors at adjacent indices satisfy (66)–(67). Search supports should be adapted to those exact affine relations, rather than treating the two pole slopes as independent tropical weights.

3. **Build or exclude deep initial-form cascades.** Cyclotomic factors provide finite residue identities, but a successful construction must combine them into multi-term faces that cancel at the same structural primes through many powers. Conversely, a resultant or height theorem proving the lower bound (98) would eliminate the whole mask architecture.

4. **Exact certificate search and Lean replay.** Contact equations, toric reduction, shared-prime congruences, valuations, and the reduced denominator are finite integer data. Directed real intervals are needed only for the final strict inequality, making independent replay and kernel verification realistic.

The conceptual lesson is now more precise than "denominator cancellation is hard." Continued fractions provide excellent near-units and Prouhet masks provide excellent contact, but the prime supports do not move: they stabilize. The scarce resource is simultaneous, deep cancellation on a fixed collection of shared structural-prime Newton faces. That is the remaining target exposed by this report.

Status: The conjecture remains open. For a fixed hypothetical candidate, one certificate satisfying theorem 9.7 and (89) would rule it out; a construction valid for every candidate would prove the conjecture. A uniform lower bound of the form (98) would instead rule out the entire exponent-lattice mask architecture.

A The general k-variable gap-ideal theorem

The bivariate theorem has a direct finite-dimensional extension useful for formalization and search.

Theorem A.1 (General augmentation-to-gap transfer)**.** *Let $F \in \mathbb{Z}[T_1, \dots, T_k]$ have degree at most d_i in T_i. Suppose*

$$F(1+W_1, \dots, 1+W_k) = \sum_{\boldsymbol{\alpha}} \gamma_{\boldsymbol{\alpha}} W_1^{\alpha_1} \cdots W_k^{\alpha_k}$$

has no nonzero term with $|\boldsymbol{\alpha}| < r$. For rational points $z_i = R_i/S_i$ with $R_i, S_i \in \mathbb{Z}$, $S_i > 0$, put $G_i = R_i - S_i$ and

$$J_F = \left(\prod_{i=1}^{k} S_i^{d_i} \right) F(z_1, \dots, z_k).$$

Then $J_F \in \mathbb{Z}$ and

$$J_F = \sum_{|\boldsymbol{\alpha}| \geq r} \gamma_{\boldsymbol{\alpha}} \prod_{i=1}^{k} G_i^{\alpha_i} S_i^{d_i - \alpha_i} \in (G_1, \dots, G_k)^r. \tag{101}$$

Consequently

$$\gcd(G_1, \dots, G_k)^r \mid J_F. \tag{102}$$

For every prime p,

$$v_p(J_F) \geq \min_{\gamma_{\boldsymbol{\alpha}} \neq 0} \left(v_p(\gamma_{\boldsymbol{\alpha}}) + \sum_{i=1}^{k} (\alpha_i v_p(G_i) + (d_i - \alpha_i) v_p(S_i)) \right). \tag{103}$$

Proof. Substitute $W_i = G_i/S_i$ into the shifted expansion and multiply by $\prod S_i^{d_i}$. Every exponent satisfies $0 \leq \alpha_i \leq d_i$, so each term is integral. The total gap degree is at least r, proving ideal membership and (102). The valuation of a sum is at least the minimum valuation of its terms, which gives (103). □

This theorem separates two phenomena. Contact controls total membership in a gap ideal; primewise cancellation can improve the tropical minimum only through ties. A proof search must therefore record both the Newton support and the residue of every tied initial form.

B Weighted contact and tropical duality

Let $\boldsymbol{\lambda} = (\lambda_1, \dots, \lambda_k) \in \mathbb{R}_{\geq 0}^k$. Define the weighted contact of a shifted mask by

$$\operatorname{ord}_{\boldsymbol{\lambda}}(F) = \min_{\gamma_{\boldsymbol{\alpha}} \neq 0} \boldsymbol{\lambda} \cdot \boldsymbol{\alpha}. \tag{104}$$

If $|z_i - 1| \leq e^{-t\lambda_i}$, then

$$|F(z_1, \dots, z_k)| \leq \|F(1+\mathbf{W})\|_1 \, e^{-t \operatorname{ord}_{\boldsymbol{\lambda}}(F)}. \tag{105}$$

At a prime $p \nmid S_1 \cdots S_k$, taking $\lambda_i = v_p(G_i)$ makes the same Newton support produce the lower bound

$$v_p(J_F) \geq \min_{\gamma_{\boldsymbol{\alpha}} \neq 0} \left(v_p(\gamma_{\boldsymbol{\alpha}}) + \boldsymbol{\lambda} \cdot \boldsymbol{\alpha} \right). \tag{106}$$

Thus the archimedean gain and the nonarchimedean floor are two evaluations of the same Newton polyhedron. Automatic termwise divisibility cannot create a contradiction by itself; the only surplus can come from different faces being active at different places, or from residue cancellation on tied faces. In the present setting, the leading-face theorem identifies the finite places at which such a change of active face is compulsory.

C Integer-programming formulation of the mask search

Fix a finite exponent support $E = \{(a_\ell, b_\ell) : 1 \le \ell \le L\} \subset \mathbb{Z}^2_{\ge 0}$ and unknown coefficients $c_\ell \in \mathbb{Z}$. Contact at least r is the homogeneous integer system

$$\sum_{\ell=1}^{L} c_\ell \binom{a_\ell}{i}\binom{b_\ell}{j} = 0 \qquad (i + j < r). \tag{107}$$

The objective $\sum |c_\ell|$ is linearized with auxiliary variables. A nonzero normalization may fix one primitive coefficient or impose a signed face sum. Toric reduction should be performed before optimization.

For a fixed adjacent candidate pair, shared-face capture is also linear in the unknown coefficients. Let $d = \max a_\ell$, $e = \max b_\ell$ and write

$$J_F = \sum_{\ell=1}^{L} c_\ell R_1^{a_\ell} R_2^{b_\ell} S_1^{d-a_\ell} S_2^{e-b_\ell}. \tag{108}$$

For every $p \in \mathcal{P}_-$ choose an allowed residual depth B_p and impose

$$J_F \equiv 0 \pmod{p^{ds_{p,1}+es_{p,2}-B_p}}. \tag{109}$$

This is a linear congruence in the c_ℓ; Chinese remaindering combines the finitely many structural primes. Taking $B_p = O(\log q)$ encodes the near-total capture forced by (97). Positivity can be enforced by a cyclotomic or sum-of-squares ansatz, or checked afterward by an exact Sturm procedure.

A practical search should not maximize contact alone. Its exact target is the scalar margin

$$\mathcal{M}(F) = r \log \frac{1}{2\Delta} - \log b_F - \log \|F(1+X, 1+Y)\|_1 . \tag{110}$$

A positive rigorously bounded margin is already a proof certificate. Negative margins reveal which prime layers dominate the failure and can guide the next face support.

D Notation crosswalk

Symbol	Meaning
x	Hypothetical nonintegral exponent.
M, N	Integer outputs $2^x, 3^x$.
ϑ	$\log 3/\log 2$.
$\nu = (a, b)$	Exponent-lattice relation vector.
δ_ν	$a \log 2 + b \log 3$.
ρ_ν, σ_ν	Input/output near-units $2^a 3^b$ and $M^a N^b$.
R_ν/S_ν	Reduced form of σ_ν.
G_ν	Gap $R_\nu - S_\nu$.
$u_{\ell,k}$	Structural valuation $q_k n_\ell - p_k m_\ell$ of the kth near-unit.
$\mathcal{P}_\pm$	Stable numerator/denominator prime supports.
$s_{\ell,k}$	Stable denominator depth $-u_{\ell,k}$ for $\ell \in \mathcal{P}_-$.
$I_{\boldsymbol{\nu}}$	Universal toric/lattice kernel of the monomial coordinate map.
$\mathrm{ord}_{\mathbf{1}}(F)$	Contact order of a mask at $(1, 1)$.
J_F	Fully cleared integer mask value.
$\rho_{\ell,k}(F)$	Uncanceled shared pole depth in the reduced denominator.
b_F	Actual reduced denominator of the rational mask value.
K_F	Compression factor from visible to reduced denominator.
$\mathfrak{c}$	Normalized denominator-compression index.
Φ_m^{hom}	Homogenized cyclotomic polynomial.

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